6 A page of printed text is placed 18 cm from a converging lens of focal length 35 cm.

Fig. 6.1 is a scale diagram of the arrangement with each of the two principal focuses (focal points) of the lens labelled F.

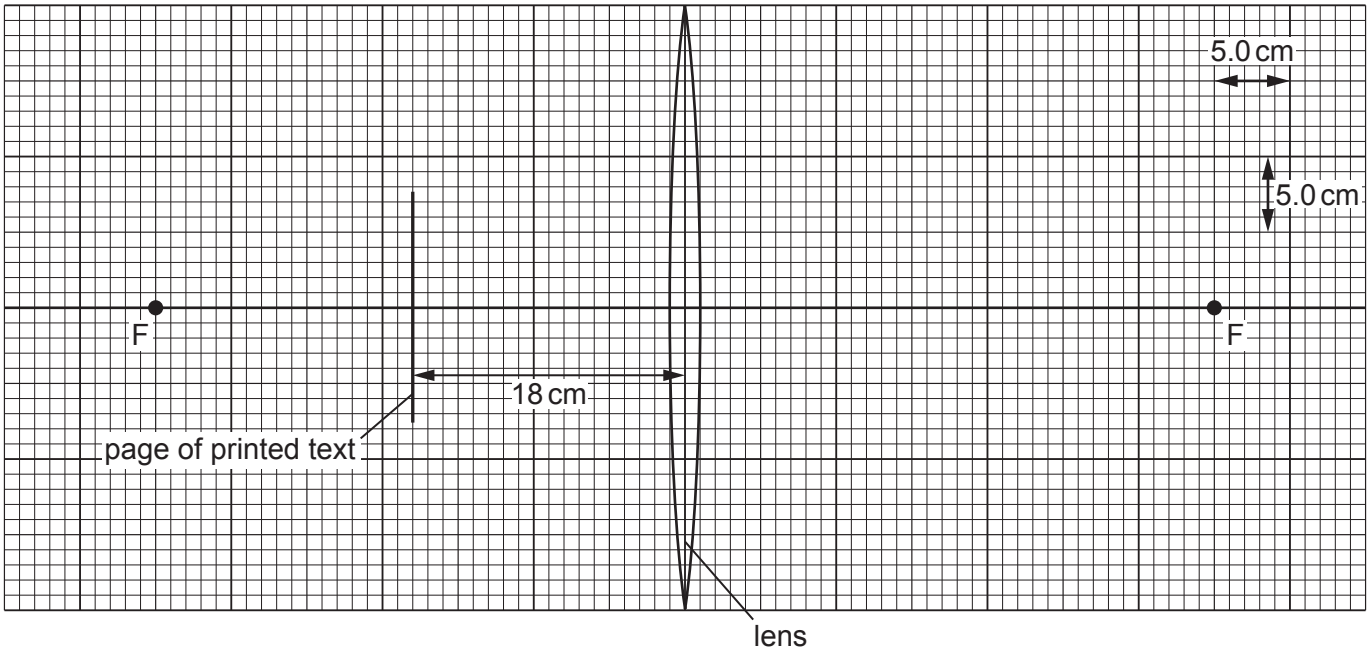


Fig. 6.1

(a) A length of 1.0 cm on the scale diagram represents an actual length of 5.0 cm.

(i) By drawing on Fig. 6.1, locate the image of the page produced by the lens and label it I. [3]

(ii) Using Fig. 6.1, determine the actual distance of image I from the lens.

actual distance from lens = ... [2]

(b) Converging lenses can be used as magnifying glasses.

State whether the image produced when a lens is used as a magnifying glass is real or virtual. Explain why.

.....

..... [1]

(c) Suggest how someone who is long-sighted may benefit from using a converging lens.

.....

.....

..... [2]

[Total: 8]